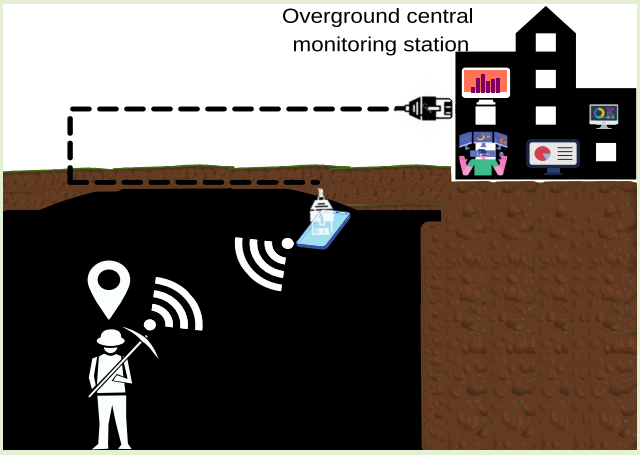
Underground Mine Positioning: A Review

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***Abstract*—Underground mining is expected to lead the mining industry in the near future. Due to the advance of electronic devices and communications technologies, new safety regulations must be accomplished in order to operate underground mines safely and optimally. In 2006, the US government updated safety policies and introduced the PUBLIC LAW109-236. As a result, underground tracking is nowadays mandatory for underground mine operations. This new regulation establishes that current, or immediately pre-accident location of all underground personnel must be delivered over ground. Despite the recent advances of tracking and positioning systems for indoor environments, underground mines are a unique scenario which poses different and highly demanding constraints for current technologies. Throughout this manuscript an extensive survey on underground mines' positioning technologies is provided. The main characteristics of current localization systems in underground mines are analysed for their implementation from a user's requirements point of view. Furthermore, the main constraints and opportunities of the existing indoor positioning systems to deliver underground tracking are presented. Finally, future research directions are highlighted.**



***Index Terms*—Indoor positioning, underground mines, industry 4.0.**

I. Introduction

THE rapid advance of wireless communications technologies as well as personal devices has allowed the deployment of different positioning systems. Positioning systems were originally created by the need of mobile users to orient in unknown environments [1]. Nowadays, the most used positioning system is the Global Positioning Systems (GPS). GPS is capable of providing a large coverage and low cost positioning for outdoor environment. This technology has been used to enable different services in areas such as transportation, gaming, health and disaster management, among others [2]. Mining industry have used this positioning technology to provide location based services (LBS) to enhance and optimize their operation. In particular, open pit (also named open cast) mining operations have been using global navigation satellite systems (GNSS) based LBS for many applications in the last 20 years. Some of these GNSS-based LBS are: drill positioning for precisely excavation of blast holes, fleet management for process optimization, collision avoidance and, transit control for safety management, among others [3]. Positioning systems in open cast mines have reached a high level of trustworthiness in the industry being an everyday technology for mine operations.

Despite the large utilization of GNSS-based LBS in open cast mining, this type of system suffers from different challenges when facing more complex mining environments, such as, deep mining pits or confined/underground areas. Satellite's signal is partially blocked in deep mining pits due to the

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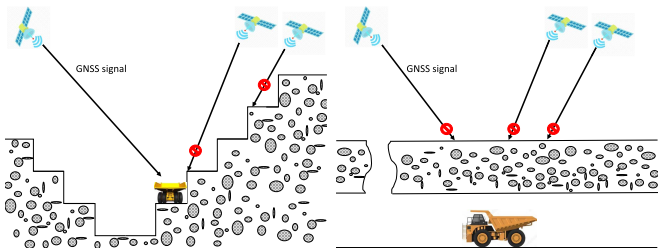


Fig. 1. Blockage of GNSS signal in (a) deep open cast mine and (b) underground mine.

presence of relatively high obstacles. Moreover, complete signal lost is experienced in underground mines and confined environments since the signal is not capable of penetrating concrete, metal or ground as shown in Fig. 1 [4].

As ore exploitation continues, shallow deposits are becoming exhausted and new veins containing high grade mineral are encountered in deeper layers of earth. Underground mining is, therefore, becoming more attractive due to their merits in terms of safety, production rates, decreasing costs and environmental friendly production [5]. Commonly, the deposits exploited by open pit mining are shallow and they have a relatively uniform geology. On the contrary, underground operation is preferred for deeper deposits which are not necessarily uniformly distributed. Due to this, underground mines are typically extensive labyrinths that employ a huge number of employees working over an area on many square kilometers [6]. In the recent years, due to the current advance of wireless technologies, new safety standards have been appointed in order to operate underground mines safely. In particular, U.S. government has changed their safety policies. In 2006, the congress introduced the PUBLIC LAW 109–236(M) [7] to amend the mine safety and health act of 1977. The main goal of this new standard is to provide wireless communications and, position information of personnel inside the mine following a underground accident.

Besides the current safety regulations for underground mines, nowadays, wireless communications and positioning systems are two key enabling technologies for industry 4.0. In particular, positioning is used for many applications in industrial environments and it can be used to provide different LBS to improve operation's efficiency and safety [8].

The implementation of such systems has encountered various challenges due to the particular characteristics of underground mines. The main objective of this article is to provide an in depth survey on underground mine positioning systems that have been proposed in literature. We focus the analysis on underground mines because they pose a more challenging scenario compared to traditional indoor environments.

A. Existing Surveys on Underground Mine Positioning

In the literature, only few surveys on localization systems for underground mines can be found. In [9] the authors briefly discuss enabling technologies to provide positioning in underground mines. In this article, the performance of the methods is not delivered. In addition to this, the main requirements of positioning systems for underground mines are not analysed. In [10] the capability of different wireless communications systems to provide positioning in underground mines are compared. In this book, a Zigbee-enabled active RFID tracking and monitoring system for miners and equipment is presented. The systems provides the position of mobile devices inside the tunnel by means of a proximity-method. In addition to this, the mobile node can alert the central station via text message in case of emergency or other requirements. Nevertheless, no exhaustive literature review can be encountered in this work and it is mainly focused on the proposed system.

In [11] different positioning technologies for underground tunnels are presented. Although in this work the authors do not deal with the particular characteristics of underground mines, a brief state-of-art analysis is done. Details of the methods and the used taxonomy for comparison are not presented. In addition to this, no performance indicators and systems requirements are discussed.

B. Key Contributions

In this paper, an intensive survey on different positioning systems for underground mines is delivered. Different systems proposed in literature are analysed in terms of their particular performance in a highly demanding and hazardous environment such as underground mines. Moreover, this work proposes a brief tutorial in localization methods focusing, mainly, in their characteristics and requirements. The main contributions of this article are summarized as follows.

1) This manuscript provides an in depth analysis of user's requirements for underground mines positioning systems based on the current legislation.
2) We address and analyse the main problems encountered for deploying localization systems in underground mines such as low or no infrastructure and harsh environmental conditions among others.
3) We analyse the main technologies proposed for underground mine positioning as well as the algorithms used for providing location information of mobile nodes inside the tunnel.
4) The main advantages and constraints of localization technologies and methods when used

C. Structure of the Paper

This paper is structured as follows: in Section II the importance of underground mine positioning is highlighted and different applications are discussed. In Section III the main requirements for underground scenarios are described based on the current legislation and environmental condition in mining tunnels. In Section IV a classification for different positioning system solutions that can be found in the literature is proposed. In Section V a bibliographic review is delivered based on the taxonomy proposed in Section IV. Finally, in Section VI the main conclusion of the study are delivered.

II. Importance, Challenges and Potentials of Positioning in Underground Mines

Underground mines can be classified in two types based on the manner it is excavated. The structure of the mining area is determined by the ore distribution [12]. If the ore

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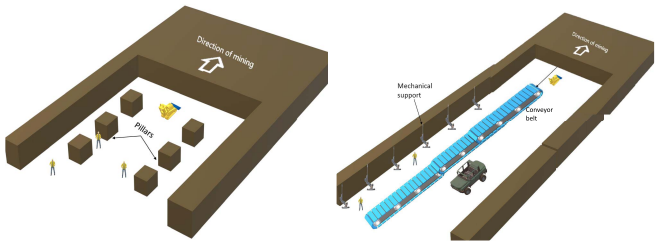


Fig. 2. Different types of mines based on their excavation (a) room-and-pillar and (b) longwall mining.

body is flat, room-and-pillar mining is the preferred method. On the other hand, when the ore body has a large, thin, seam-type shape, longwall mining is used. The first type, room-and-pillar mining, is a cyclical and step-by-step mining sequence in which, seams of mineral are mined partially, leaving large pillars of mineral while excavating in order to support the roof from possible collapse [13]. This method creates a network of alternating open spaces and large pillars of coal as shown in Fig. 2 (a). Once the ore deposit has been exhausted, the remaining pillars are removed and the roof is allowed to collapse. This practice is particularly dangerous and is a major and is a major cause of accident inside the mine operation [14]. On the other hand, longwall mining consist on the excavation of a non-interrupted and continuous tunnel using a shearer as shown in Fig. 2 (b). Hydraulic supports must use for supporting the mine's roof at the longwall face [12]. The type of method used for mine excavation has an impact on the localization technology used to determine the position of a mobile node. The tunnel's shape will affect the signal propagation characteristics. In [15] different cases are consider in order to model the propagation characteristics of RF signals in underground mine environments. In particular, straight tunnels with LOS conditions between transmitter and receivers and near face scenarios are examined. It was shown that near to the working face the coverage and data packets reception probability are reduced compared to straight tunnel scenario.

Underground mine environment is dynamic by nature. As ore exploration continues, mine expands. Therefore, it can be considered as a semi-structured environment [16] where some assumptions on the geometry of the tunnel can be made. Moreover, due to the characteristics of the mining tunnels, low control on the localization infrastructure exist and strategic deployment of required devices for positioning cannot always be done. From a positioning system perspective, semi-structured environments pose huge challenges in terms of system scalability (to expand the coverage of localization system) and precision. In addition to this, the evolution of the tunnel must be followed closely in order to update the maps of the mine site. Underground mines environmental conditions are considered by several researchers as one of the most difficult scenarios [17]. Underground mining operations involve various hazardous conditions, such as, in-mine vehicular/human accidents, fire and explosions, collapses, toxic gases emanation and floods, among others. In addition to this, workers, machinery and in-mine equipment are exposed to extreme conditions, such as, high levels of humidity (up to 90 % or above), airborne dust and extreme heat. These conditions have not only an impact on workers performing labours inside the tunnel but also on the functioning of electronic devices. Due to the above described conditions of underground operations, accidents are more dangerous than those in other activities. Mine operation, as any other industrial labour, tries to achieve maximum productivity by working safely. As a result, positioning systems must be designed to fulfill their special requirements and, overcome a huge number of challenges in terms of design and implementation. In [17] different considerations for underground communication devices, such as, safety, immunity and protection, size, design and reliability are presented. The analysis of these requirements is done by taking into account the extreme conditions in which the devices must operate when they are used in underground environments. Positioning technology for underground mines must be designed taking into account the same special requirements of communications devices since they will be exposed to the same conditions. Underground mine positioning systems can not only be used for safety enhancement purposes, such as employee condition and environment monitoring systems, post-accident rescue coordination and accident investigation and refuge-alternative navigation system, but also to improve the mine operation by enabling mining automation and remote operation, which is fundamental in nowadays underground mines [18].

III. Requirements of Underground Mine Positioning Systems

Indoor positioning is often viewed as an multidimensional optimization problem where there exist a tradeoff between its characteristics. In this section, different positioning systems requirements are detailed and analysed specifically for underground mines. The UM-IPS requirements are organized in such way that first, market related characteristics are described. These are: Market maturity, approval and cost. Then, requirements related to the necessary infrastructure, service availability and intrusiveness are discussed. Here, requirements such as, infrastructure, intrusiveness, coverage, scalability and number of users are analysed. Finally, position determination requirements are described for underground mines. In this group, characteristics, such as, accuracy, precision, update rate, output data and robustness will be examined.

A. Market Maturity

The market maturity states the current development of the proposed technology. Different maturity levels can be identified to classify the positioning system based on the work done by Mautz [19]. In particular there are three main market maturity levels

- Concept: Concept is the early stage on the development of an indoor positioning system. In this stage simulations, as well as, evaluation of different system's aspects are carried out. Requirements analysis is expected to be done in this stage.

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- Prototype: A second step in the development of a positioning system is the implementation of a prototype. In this stage of the development a real platform in which the system is tested must be provided. Performance of the prototype should be tested using lab conditions, a simulated environment or real conditions. The latter, provides the most useful information about the positioning system and is often identified as a necessary step to move on and advance to the next stage.
- Product: Is the final stage in a positioning system development. In this stage, the proposed system is available in the market.

In general, mining industry is opposed to the changes. Due to this, only product level technologies are preferred when implementing positioning systems for underground mines.

B. Approval

Some regulations establish the need of approval from governmental or private agencies for the implementation of positioning systems in some environments. As an example, the MINERS-Act of 2006 [7] establishes the need of approval from Mine Safety and Health Administration (MISHA) in order to be capable of functioning in an underground mine.

C. Cost

The cost of a positioning system can be divided in different sub-categories where not only monetary price is considered.

- Time cost: considers the time required to install and administrate the positioning system.
- Capital cost: The capital cost considers the infrastructure cost (prices per mobile and static devices) and salaries of support personnel.
- Maintenance cost: Is the cost for maintain the system functioning.
- Space cost: Takes into account the amount of required infrastructure and hardware's size.

In particular for underground the cost must be the lowest possible in these three categories. Low maintenance cost will ensure continuous operation and will increase the safety. A low capital cost will allow operation to implement these systems massively and a low space cost is required due to hazardous conditions of underground mines and its lacks of space for deploying infrastructure.

D. Required Infrastructure

The required infrastructure refers to the deployed hardware needed to perform positioning. The required infrastructure may vary depending on the localization solution. As an example, no infrastructure is required for pedestrian dead reckoning (PDR). On the other hand, dedicated infrastructure must be deployed for radio-frequency identification (RFID) and some Bluetooth based solutions. In these applications hardware is deployed in order to provide exclusively location services. Finally, WiFi and visible light communications (VLC) based positioning solutions are considered as pre-deployed infrastructure since they use the already deployed network to provide communication and positioning. Although lighting and WiFi network deployments are present at some extent in underground mines [20], [21], due to harsh conditions it is not always possible to ensure that the entire mine site will be covered by these infrastructure deployed networks. Due to the particular characteristics of underground mines, positioning system which requires minimum or none infrastructure are preferred.

E. Intrusiveness

Intrusiveness of a positioning system is refereed to the act of interrupting and disturbing. The positioning system is generally composed by one or several devices. These devices are installed in mobile nodes or as deployed infrastructure. Due to the particular characteristics of underground mines, the fixed infrastructure deployed inside the tunnel must be as low as possible. It means that this infrastructure must be imperceptible. On the other hand, intrusiveness of mobile devices must comply some other requirements. In Mardonova et.al. [22] different wearable devices are analysed for its usage in underground mines. Is important to notice that the mobile device attached to the miner which is used for positioning must be imperceptible for performing labours safely.

F. Coverage

The coverage of a positioning system can be defined as the spatial extension in which the performance of the system must be guaranteed [19]. Three categories of coverage are available:

- Local coverage: Local coverage is refereed to a small and well-defined area which could be a tunnel section, gallery and stope, among other mine sections. Local coverage is not extendable (or scalable). Its coverage is defined in m, m^2 or m^3.
- Scalable coverage: This type of systems have the ability of extend the area of coverage by adding hardware infrastructure. In addition to this, the system must not compromise its accuracy when extending its coverage.
- Global coverage: It is a system that can perform worldwide. Only GNSS system and celestial navigation can be considered in this category.

Underground mines are a dynamic environment. Due to this, scalable coverage is preferred since new tunnel sections are being open and close as the operation continues.

G. Scalability

The scalability refers to the ability of the positioning system to increase its coverage area. The positioning system is not scalable if it cannot provide position estimation in a larger area than the originally designed. On the other hand, scalable systems can increase its coverage area. For doing this, increment of required infrastructure may be needed. Moreover, loss of accuracy could be experienced as side effect of increasing the coverage. Underground mines are known to be a dynamic environments, since working faces and transport tunnels are continuously being created to reach novel mineral deposits or closed when the valuable mineral resources are exhausted. Due to this, the proposed positioning system for underground mines must be scalable and should adapt to mine transformations.

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H. Number of Users

The number of user is often based on the type of user the systems can support. In underground mines, position information of all workers inside the tunnel must be available for the overground monitoring station. Regular workers, visitors and people inside vehicles and machinery must be identified and localized.

I. Accuracy

Accuracy of a localization system can be defined as the deviation between position estimation and true value. The concept "measurement accuracy" cannot quantified numerically. A position estimation is said to be "more" accurate when it offers a smaller estimation error. In order to classify methods by its accuracy, statistical information of estimation error is used. Typically, positioning error is measured by taking the euclidean distance between real and estimated position. Accuracy of a system is often obtained by using the cumulative distribution function (CDF) of positioning errors. Some other positioning systems provide the mean value of errors in order to provides accuracy estimation. High precision accuracy is not required by MINERS act in underground mines since tunnel may have hundred kilometers length. Fine grained positioning is hard to be accomplished in the entire tunnel and accuracy requirements can be heterogeneous depending on the activities taking place in particular tunnel sections. In the Miners-ACT [7] an accuracy of 60.96 meters is established as the minimum accuracy achievable by the positioning system.

J. Precision

Precision measures the deviation of position estimates for the same location. Precision and accuracy are often assumed as the same concept in real life situations. However, there are fundamental differences between the two concepts. Precision is often related with the standard deviation of consecutive measurements under the same conditions. This parameter is often not defined for indoor positioning systems. In particular, for underground mines this characteristic is not specified by any standard.

K. Update Rate

The update rate is the maximum time in which the position of mobile node must be refreshed. Traditionally, real time localization services for pedestrian usage have an update rate of 1 Hz (position estimation each second). This parameter strongly depends on the mobile node's speed. For example, a vehicle driving at 16 km per hour travels 16.7 meters in a second. Due to this, when faster mobile node's must be localized, higher frequency update rate is required in order to ensure positioning with higher accuracy. In particular, for underground mines the update rate is set by safety regulations as shown in Table I.

L. Output Data

On specific requirements of the positioning system, different types of data can be used to represent the position of the mobile node within an environment. The most used data position representation types are 2-D and 3-D coordinates. These coordinates can be delivered using a relative or absolute positioning. In the first, coordinates are expressed as a function of an arbitrary point within the space. The second, presents the position coordinates based on an absolute reference system. Some other information can be added. For example, velocity and orientation are the data output required for IMU based positioning systems. Some other less common data output for positioning systems are uncertainty and variance. In underground mines mostly 2-D coordinates is required.

TABLE I
MINERS TRACKING SYSTEMS REQUIREMENTS BASED ON MINERS ACT [7]

Parameter	Requirement
Market maturity	Product
Approval	Requires approval from local governmental institutions (MSHA, in USA)
Cost	Low
Required infrastructure	Ambient without conditions to allow infrastructure deployment
Intrusiveness	Non intrusive
Coverage	Mine wide and scalable
Scalability	Yes, with possible loss of accuracy
Number of users	Entire staff inside tunnel, including visitors
Accuracy	60.96 m relative to a fixed point
Precision	Not defined
Update Rate	Up to 60 seconds
Output data	2-D coordinates,absolute positioning
Robustness	Depending on the technology
Privacy	Position available in a central monitoring station

M. Robustness

The robustness of a positioning system refers to its resistance against perturbations. This perturbation can be of various types. For example resistance to physical damage, jamming, dust and heat can be considered for robustness analysis. In particular, for underground mines several perturbations can be pointed out as the most significant.Depending on the technology used to provide positioning, the perturbations may differ. Due to this, robustness concept in underground mines is strongly related with the technology used to provide localization. For instance, humidity, tunnel shape and material will have an effect on RF signal [23]. Dust in suspension may affect the light propagation and continuous operation may generate noise that will interfere with acoustic signals among others.

N. Privacy

Privacy is nowadays one of the major concerns when implementing localization in industrial environments. In order to properly adopt positioning technologies in underground mines, a large number of issues concerning the capture, storage and usage of worker's positioning data must be solved. It is fundamental to ensure that the data gathered will have a proper usage. Moreover, worker's unions often argue that real time tracking could be used in an improper way. In order mitigate this problem, different method to prevent the user from being

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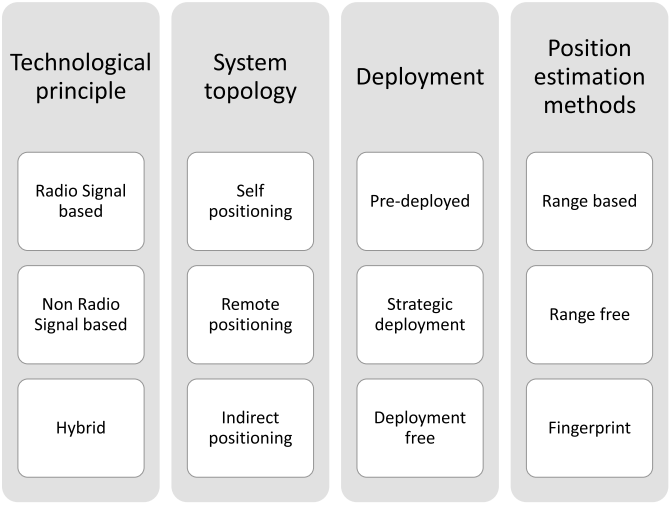


Fig. 3. Proposed taxonomy for the analysis of positioning systems.

identified have been proposed [24], [25]. In underground mines, worker's position and identification must be available in a central monitoring station located overground.

IV. Classification of Positioning Systems for Underground Mines

Positioning systems are used to determine the position of a person and/or device relative to a known position or within a coordinate system [26]. Basically, any Positioning system has two main components, i.e., "known" nodes and "unknown" nodes. Known nodes are aware of their position. On the other hand, position of unknown nodes is to be determined by the positioning system. Known nodes are also named as anchor nodes, reference nodes or beacon nodes in the literature whilst unknown nodes are often named as dumb nodes, target node or to-be-located nodes [27].

In this section, some basic working principles and requirements of UM positioning systems will be deliver. First, different requirements as well as the most important performance indicators will be explained in detail. Hereinafter, positioning systems that can be found in literature are detailed for the reader. In order to classify them, we proposed a taxonomy shown in Fig. 3.

The proposed taxonomy classifies positioning systems by using four different characteristics. Each element of the proposed taxonomy is explained in detail below.

A. Technological Principle

For classifying different types of localization systems based on their technological principle, three main categories are proposed. The first category covers the entire range of systems using radio frequency signals to determine the position of mobile nodes. On the other hand, non-radio signal based systems cover the spectrum of system that does not require a wireless-radio based network to determine the position of the mobile node. A third technological approach are hybrid systems who use both technologies to provide localization. The three types of technologies that are described in detail bellow

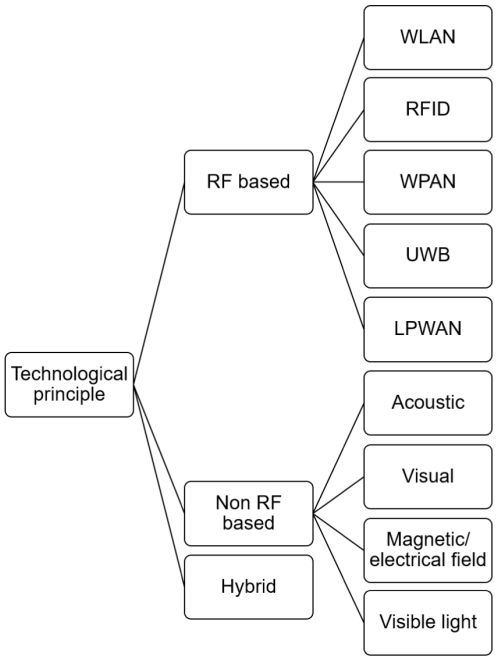


Fig. 4. Classification based on technological principle.

based on research conducted by several authors. In addition to this, to summarize the analysis a table of different localization systems proposed in literature is presented at the end of this section.

1) Radio Signal Based Systems: Radio signal based systems are those that use the radio-frequency spectrum of the electromagnetic waves to determine the position of the mobile node. Radio signal based systems has the advantage that they can be used to provide two main pillars of location based services, i.e., help in the position estimation process and the capability to transmit wireless information allowing mobility of the unknown node. Underground mines are a very challenging environment for radio frequency based signals. Behaviour of RF signals is different in mine tunnels than in any other regular environment. This, due to the physical shape of the tunnel can cause signal scattering, multipath propagation and blockage (pillars, U turns, etc.). Moreover, tunnel walls are composed by different materials depending on the mine site. Due to this, electromagnetic characteristics such as dielectricity and conductivity of each tunnel are unique and they have an effect on the radio frequency signal propagation. The proposed radio signal based system must cope with these challenges.

- Wireless Local Area Networks (WLAN): WLAN (IEEE 802.11) is a well establish wireless communications technology in various industry sectors. WLAN standards use the Industrial, Scientific and Medical (ISM) bands in 5 and 2.4 GHz to provide full-duplex wireless communications. This type of networks are thought to deliver data transmission within a rate between 1 to 108 Mbps and a coverage range between 50 and 150 meters depending on the characteristic of the environment and line-of-sight (LOS) is not required for data transmission. Experimental demonstration of WLAN, in particular IEEE 802.11b standard has been conducted in [17]. The mine site

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where WLAN was tested correspond to a longwall tunnel. Ranges up to 150 meters were achieved, whereas 802.11b devices are expected to provide a maximum range of 50 meters. In case WLAN is provided as already-deployed communications infrastructure, localization systems can be rapidly delivered without requiring any further deployment. WLAN standards provide the capability to implement diverse position estimation methods. Particularly, proximity and lateration methods are preferred since off-the-shelf devices are equipped with the required hardware.

- Radio Frequency Identification (RFID): This type of systems are capable of store information and retrieve data using radio waves. RFID systems comprise two main components i.e., RFID reader and RFID tags. RFID reader is capable of obtaining the information stored in RFID tags using a pre-defined protocol. RFID tags can be classified in three groups, these are: passive tags, semi passive tags and active tags. Active tags transmit their information to reader previously to a query or independently. Since active tags are equipped with batteries they are capable of working with low-power received signals. In addition to this, they can be associated easily with different type of sensors. Nevertheless, due to the usage of batteries, they are more expensive and have a more limited lifetime compared to passive and semi passive tags. Semi passive tags are known also as battery-assisted tags use their battery prior to a reader query in order to increase its communication range. They have a battery life of about five years and once it is drown they acts as passive tags. Their coverage range is between active and passive tags ($<$10 meters). Passive RFID tags can operate without a battery. The RF signal coming from the reader power ups an integrated circuit that generates the response to the reader's query. Their main drawback is its short communication range (3-5 meters). Usually, RFID reader and tags are not pre-installed as communication infrastructure. Due to this, this type of system needs to be deployed for a specific purpose. Moreover, due to its short coverage range, strategic deployment of readers is required in order to provide full coverage.
- Wireless personal area networks (WPAN): WPAN, standardized in IEEE 802.15 have been extensively investigated in underground environments. These networks offer bi-directional wireless communications. Compared to WLAN their coverage range is much shorter and they do not possess pre-deployed infrastructure. Standards such as IEEE 802.15.4a and ECMA-368 (which will be discussed next) include positioning features. Nevertheless, they have not been used extensively in positioning systems intended for underground mines. In particular, two standards are preferred for using in underground scenarios, i.e., Bluetooth (IEEE 802.15.1) and ZigBee (IEEE 802.15.4). Each WPAN standard differs significantly in their physical layer. Due to this, each standard will be detailed separately below.
  |*Bluetooth*: This type of systems have a maximum coverage range of up to 10 m depending on the propagation conditions of the environment when a class 3 transmitter with a power output of 1 mW are used. They use the ISM band in 2.4 GHz to provide wireless communication with a rate between 1 and 3 Mbps. In general, off-the-shelf bluetooth devices are not equipped with the necessary hardware to provide lateration-based positioning. Due to this, proximity methods are preferred in this type of systems since no extra hardware is required.
  |*ZigBee*: ZigBee technology is a low data rate WPAN standard with data rate up to 250 Kbps operating in the band of 2.4 GHz. This standard is also specified for 868–870-MHz and 902–928-MHz bands at 20 and 40 kbps, respectively. ZigBee networks are principally intended for low cost and low power monitoring and control applications. The coverage range of ZigBee nodes in indoor environments is 20 to 30 meters. ZigBee technology is capable of providing lateration and cell of origin based positioning using off-the-shelf devices.
- Ultra Wide Band (UWB): Ultra wide band systems have been subject of extensive research as a promising candidate for WPAN applications, sensor networks and ubiquitous computing [28]. Due to its large bandwidth usage, regulation of these type of systems required a completely different approach. For instance, frequency channels assignation of such large widths to specific users are not feasible because free spectrum is not available. For instance, ECMA-368 standard for WPAN uses the frequency band of 3 to 10 GHz. IEEE 802.15.4a include for the first time UWB and designated operating frequencies in three ranges: bellow 1GHz, between 3 and 5 GHz, and between 6 and 10 GHz. UWB technology has extensively adopted for localization systems where accuracy on the order of 1 meter is required. Its high accuracy is due to the precise time-of-flight range estimation achieved by its large bandwidth. IEEE 802.15.4a is an example of a standard capable of providing ranging accuracy of 1 meter or better.
- Low Power Wide Area Network (LPWAN): low power wide area network is a wide-range communication technology with low power consumption. This technology has been pointed as one of major enablers for expanding internet of things in large-scale urban areas. In recent years, several standardized and proprietary LPWAN solutions have emerged, such as, IEEE 802.15.4g Long-Range wide area network (LoRaWAN), Sigfox, Ingenu, Weightless and Narrow-Band IoT (NB-IoT), among others. Due to their large coverage they have been mostly used for large scale indoor/outdoor positioning or complement GPS in urban areas, where satellites signals are prompted to be blocked by high buildings. The potential for their usage in indoor positioning systems have been recognized and analyzed in [29]. Here, the potential of using different IoT based solutions for indoor positioning was analysed. Up to date, some efforts have been done to evaluate this technology in GPS-denied scenarios, such as, vehicular tunnels [30]. As far of author's knowledge, this technology has never been applied to provide underground mine positioning and only few link performance measurements

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and environmental monitoring applications has been proposed [31], [32]. Due to their different characteristics, three of the major LPWAN solutions will be detailed separately below. These technologies were chosen among the others because they use unlicensed spectrum for their implementation.

—*LoRaWAN*: This LPWAN is based on LoRa modulation, which is a proprietary radio modulation scheme based on Chirp Spread Spectrum (CSS) designed for IoT devices. LoRa modulation provides resilience against interference, multipath and Doppler effects. This technology uses the sub-GHz spectrum (868 MHz, 915 MHz and 433 MhZ). The operating frequency of LoRa devices depends on local regulations, as for example, 868 MHz is used in Europe and 433 MHz in Asia. Working in sub-GHz frequency range allows LoRaWAN to be more robust to line-of-sight obstructions and increase their coverage. The above mentioned characteristics could be useful in underground mines where the presence of large moving machinery, static obstacles and tunnel curves may affect RF signal propagation performance. LoRaWAN can provide data rates ranging from 3 to 50 kbps depending on spreading factor and channel bandwidth. Furthermore, this type of networks can achieve a coverage between 5 to 20 km in urban/rural scenarios, which is a promising feature for underground tunnels. LoRaWAN is an open standard and this type of networks can be implemented without paying fees or asking for permission to a centralized authority. This reduces the cost of their implementation which could meet the cost requirements of UM-positioning systems.

—*Sigfox* This technology has experienced rapid development and it has been implemented over 80 countries [33]. There exist a major difference between Sigfox networks and LoRaWAN, this is that Sigfox company acts itself as the network operator. The power consumption of the Sigfox mobile node is lower compared to LoRa devices at expense of decreasing the achivable data rate to 600 bps. This characteristic also produce a cost reduction. Sigfox uses ultra narrow band (UNB) technology and allows users to select chipsets from various chip manufacturers (contrary to the case of LoRaWAN, where patented LoRa chipset must be used). One of the major problems for their implementation in underground mines is that register and payment to the Sigfox company is required. Moreover, the data has to be stored on the Sigfox server, which could create privacy problems.

—*HaLow*:IEEE 802.11ah or HaLow is a wireless protocol that uses 900 MHz license-exempt bands. The main objective of this type of networks is to extend the coverage of traditional WiFi networks which operates in the 2.4 GHz and 5 GHz bands. Its coverage range is limited compared to LoRaWAN and Sigfox networks reaching up to 1 Km range. On the other hand, it can achieve data rates from 150 Kbps up to 347 Mbps. Due to its shorter range the infrastructure required to provide positioning in underground mines is larger compared to LoRaWAN and Sifox. Nonetheless, a major advantage of WiFi HaLow is that it does not require a proprietary hardware and service subscriptions. Off-the-shelf IEEE 802.11ah routers can be used to provide wireless communications and underground mine positioning.

2) Non-Radio Signal Based Systems:

- Acoustic systems: In contrast to electromagnetic waves used by RF systems, acoustic signals are mechanical waves. Mechanical waves are transmitted by the oscillation pressure in a medium such as air or building materials. Acoustic position estimation can be based on ultrasound or audible sound. The estimated range of acoustic based positioning systems is about 10 meters due to the specific decay profile of the airborne acoustic channel. Lateration based methods can be easily implemented using this type of systems using a microphone array at the receiver side. Acoustic systems are severely affected by multipath propagation, directivity of transducers and noise. In addition to this, increasing the power of the signal is not a feasible alternative since loud sound signals from nodes to be localized are energy-demanding which decreases the battery lifetime. Furthermore, the presence of machinery and continuous mine operation may affect the acoustic system due to interference and saturation of the receiver microphone.
- Inertial Navigation Systems (INS): they estimate the mobile node position, velocity and orientation from an Inertial Measurement Unit(IMU). The inertial measurement unit consist of three motion sensors (orthogonally arranged accelerometers), three angular rate sensors Gyroscopes) and/or a magnetometer (measuring the strength and/or direction of a magnetic field). This type of systems provides independent operability. Due to this, they do not required a deployed infrastructure to estimate the mobile node position. Accelerometer measurements are also corrupted by noise. Due to this, INS decrease their accuracy over the travelled distance. Finally, magnetic fields inside the mine are inhomogeneous and dynamic due to the presence of machinery. Therefore, magnetometers measurements could be corrupted and affect the overall performance of the position estimation.
- Signage and Maps: Signage and maps are traditionally the most available approach for human navigation in indoor environments. Actual position is provided by using static signalling and, in order to help the mobile user to reach his destination, a partial or complete map display of the environment is provided. Nowadays, new signs and maps are replacing this traditional approach. For instance, cameras and mobile devices can replace user's sight and reasoning to provide positioning and navigation. In [34] the usage of a camera to scan unique "markers" is proposed. By doing so, the unique mark provides information to a mobile phone application to determine actual position. On the other hand, the approach proposed in [35] tries to match the complete acquired images with a database of the building's floor plan. If a match is found, the current location is displayed.
- Magnetic/electrical field: Contrary to electromagnetic waves (RF systems), magnetic waves are capable of

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penetrating walls inside buildings without propagation error. Due to this, magnetic field based localization systems does not suffer from several drawbacks encountered in electromagnetic systems, such as, multipath and signal delay effects among others. When using magnetic fields for indoor positioning, a distinction on the nature of the magnetic field must be done. Two different types of systems can be considered, natural magnetic field based system and artificial magnetic field based system. Additionally, natural can be classified in two sub-types, geomagnetism and permanent magnets. Geomagnetism based systems use the natural earth's magnetic field to provide positioning. Inherent structure on interior tunnels produce disturbances on the natural field. These variations can be used as a fingerprint to determine the mobile node position. On the other hand, permanent magnets can be used in the mobile node. Magnetic field sensors place at known places measure the magnetic density flux produced by the magnet in order to determine its position. The second type of localization systems based on magnetic fields use artificially generated magnetic fields. In this type of systems, a variable electrical current circulating in concentric coils is used to create the artificial and controlled magnetic field (Lenz effect). Usually, the artificially induced magnetic fields have a short coverage range (less than 3 m) and, due to this, their application is limited to small volumes.

- Visible light communications (VLC): These systems use the visible spectrum of electromagnetic waves to provide communication. This type of systems have recently emerged and has been subject of extensive research for the research community. The recent advances in lighting technology, in particular lighting emitting diodes (LED) have transformed VLC system in a feasible and applicable technology. VLC technology transmits information by modulating the intensity of an optical source at a rate much more faster than the response time of human's eye making the change on the optical source unnoticeable. As a consequence, VLC systems are capable of providing illumination and wireless communication at the same time. As the lighting infrastructure is assume to be predeployed, only a few modification on the lighting devices is required. These systems used require mostly line-of-sight (LOS). In addition to this, VLC link is used mostly to provide the downlink and, hybridization with RF and Infra-red (IR) systems is proposed to provide full-duplex communication. In literature, the term visible light positioning (VLP) stands for localization systems that use VLC to provide positioning. In general, many of the approaches that used in RF based positioning systems, such as, range free, range based and fingerprinting method. The propagation of light signal could be affected by several factors when facing working face conditions. Dust in suspension produce scattering of the light signal and deteriorate link performance. Moreover, partial of total LOS lost could be experience due to large mobile and static machinery present in underground tunnels.

3) Hybrid Systems: as it can be seen, numerous technologies can be used to provide positioning. Each technology possess their own *pros* and *cons* in terms of performance, coverage, etc. Due to this, the idea of combining different types of localization technologies to overcome difficulties and/or improve the system's performance arises. The main problem of this type of systems is that there does not exist a unified systematic evaluation on the possible trade-offs and drawbacks of using different technologies simultaneously for localization purposes. In the literature, diverse hybrid localization systems be encountered. They are mainly focused in increase the performance (mainly accuracy) and coverage of traditional single-technology approaches.

B. System Topology

The deployment of anchor nodes as well as the mobility of dumb nodes has an important effect on the performance of the localization system. [85]. Localization systems can be classified in three different groups considering the topology of the system, i.e., self-positioning, remote-positioning and indirect-positioning. This classification is useful when evaluating a positioning system for a particular application. Different approaches will be classified based on where the position measurements are performed and where the obtained position information is used. [85]. The three categories are detailed below.

- Remote positioning: In this type of systems the anchor nodes receive the signal originated from or reflecting off the dumb node. Measurements of the signal are transmitted to a central station where they are used to estimate a position as shown in Fig.5(a). These type of positioning systems are used in centralized LBS.
- Self Positioning: In self positioning systems the dumb node performs the measurement of the signal sent by the anchor nodes and uses these measurements to perform localization. Using this type of topology the mobile node or user is aware of its position. Nevertheless, this position information can be only used in distributed applications as shown in Fig.5(b).
- Indirect Positioning: When the networks used to perform localization is capable of transmitting data indirect positioning topology can be used. In this scheme the estimated position by self-positioning or remote-positioning can be re-transmitted as illustrated in Fig.5(c). When the self-positioning system sends its information to a remote or central station is referred to as indirect remote-positioning. On the other hand, when a remote positioning system delivers to the dumb node its positioning information is referred to as indirect self-positioning.

As it can be seen from Fig.5 the system topology will impose a major constraint on the LBS that can be provided. For example, navigation systems are thought to be enabled by self-positioning systems since the mobile node must be aware of its position in order to be guided. On the other hand, dispatch software use remote-positioning topology since they require position information of every dumb node in order to optimize the instructions The topology that fulfill

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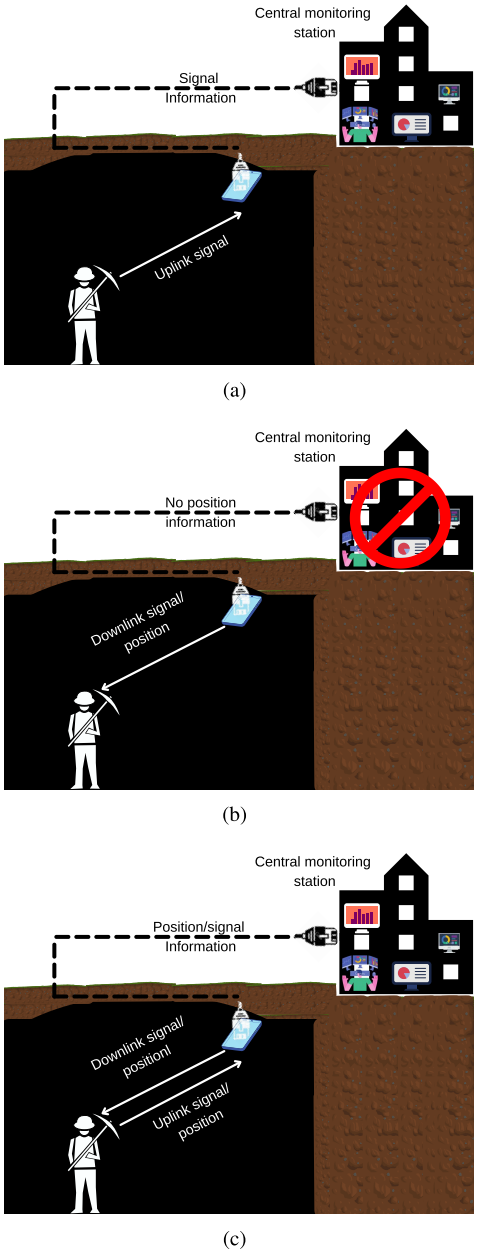


Fig. 5. Different network-based positioning methods (a) Range based, (b) Range free and (c) Fingerprinting.

both requirements is the indirect position where the location is known in both sides (central monitoring station and the mobile node). In the following section, different enabling technologies are analysed. Their potential to provide self-positioning, remote-positioning or indirect positioning will be delivered as part of this investigation. The technology that enabled indirect positioning will have an advantage over the others since it could enable a larger number of applications.

C. Deployment

This characteristic of the localization system defines the deployment of required infrastructure to provide positioning. The infrastructure deployment is classified in three categories detailed below

- Pre-Deployed: In this type of localization systems the required infrastructure is already deployed in the scenario. In particular, systems that uses the wireless communication infrastructure (WLAN, VLC, etc.) to provide localization are the most representative systems.
- Strategic Deployment: This type of systems require special deployment of localization infrastructure to provide positioning. Systems such as RFID, magnetic, acoustics and WPAN based systems are some examples of localization technologies that require the installation of dedicated infrastructure to determine the position of the mobile node.
- Deployment-Free: These systems do not require infrastructure deployment to provide positioning. Usually, inertial sensors and map knowledge are used in indoor scenarios.

D. Position Estimation Method

Positioning methods based on network deployment can be classified in three types [86]. This classification is based on whether they require a distance (range) estimation or some other signal information to compute mobile node's position [19]. These three main positioning approaches are: (1) range-based methods, (2) range-free methods, and (3) fingerprinting which are shown in Fig. 6. Another important family positioning methods that can estimate user's position without a network deployment are dead reckoning method which are classified in a completely different group (4).

1) Range Based Methods: These algorithms determine the mobile node position by using a distance estimation. The three main range based methods are lateration, hyperbolic lateration and angulation. The before mentioned methods require a distance estimation to perform localization. This estimation is derived from signal characteristics, such as, received signal strength, time of propagation, and received phase/angle. Receiver signal strength (RSS) based ranging is the simplest method to derive distance estimation. RSS measurements are relatively easy to obtain and, most of the devices are built with the required hardware to measure received signal power level in dBm or a received signal strength indicator (RSSI). In order to estimate the distance from RSS measurements, a precise model of signal propagation is required. Due to this, model precision has high impact in algorithm performance. On the other hand, despite provide a more precise distance estimation than RSS methods, time of propagation and received phase/angle measurements have several drawbacks on their hardware implementation. Time of propagation measurement can be divided in time of arrival (TOA) and time difference of arrival (TDOA) methods. TOA measurements require precise synchronization of transmitters and receivers which is very difficult to provide. On the other hand, TDOA and phase/angle measurements require synchronization at the receiver side to estimate the range. In addition to this, to perceive the difference in time of arrival, receivers must be equipped with a high sampling frequency device when technologies such as RF and VLC are being used.

Once range measurements are performed, range based methods estimate the position of the mobile node by

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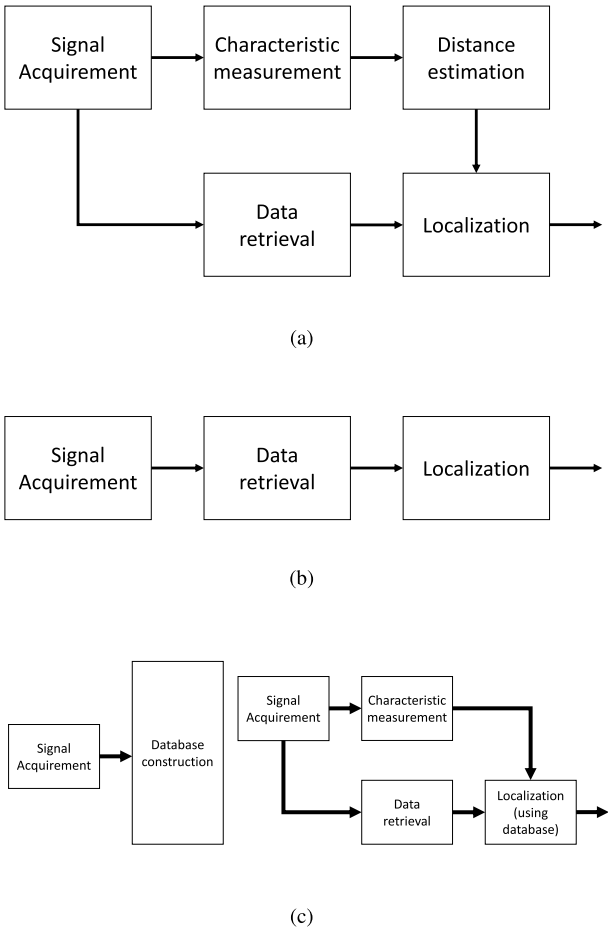


Fig. 6. Different localization system topologies (a) Remote positioning, (b) Self positioning and (c) Indirect Positioning.

trilateration/multilateration/angulation algorithms. For doing this, information of multiple anchor nodes (position-aware transmitters) is required. This, imposes another drawback on the implementation of type of algorithms.

2) Range Free Methods: Range free algorithms use only connectivity information to perform localization. Different techniques such as, equalization and coding can be applied in order to successfully deliver the message from beacon node to the mobile user. Due to this, signal perturbations do not severely affect position estimation [27]. Moreover, no extra hardware is required to measure signal characteristics. As a result, they provide a more robust approach compared to range based algorithms [87]. Nevertheless, there exist a trade-off between its high robustness, low hardware complexity and their achievable accuracy. Range free algorithms provide coarse-grained localization (low accuracy) whilst range based algorithms can provide a fine-grained localization (high accuracy).

Traditional range free algorithms mainly used in wireless sensor networks (WSN) are Cell ID [81], convex position estimation (CPE) [87] and centroid algorithm [88].

3) Fingerprinting (FP): Fingerprinting (FP), also known as scene analysis, estimates the position of mobile node by comparing the characteristics of the received signal with a previously collected fingerprint map. This algorithm requires two phases: First, an offline phase during where scene information is collected and, second, an online phase where receiver seeks the best match with the offline measurement in order to estimate its position.

Finding the best match can be carried out through probabilistic inference, or classification algorithms such as k-nearest neighbors, support vector machine, Naives-Bayes classifier, and artificial neural networks [89]. Fingerprinting method does not require LOS from different sources, unlike range based solutions.

In general, positioning systems based in fingerprinting use a type of RSS measurement to estimate mobile node's position. This measurement could be the signal impulse response [90], and power spectral density (extracted power of each modulated frequency), amon others.

FP methods do not require to receive information from several beacon LED nodes unlike range-based techniques. However, constructing the required database in the offline phase, as well as, updating the map as the configuration of the environment changes is a major challenge for this type of methods. The requirement of updated fingerprint maps increases the cost of maintenance. Moreover, FP methods performance could be affected by the blockage from bodies or new objects in the scene [89], [90]. To address this problem, in Kail *et al.* [91] a Bayesian signal model that considers the possibility of obstructions is proposed.

4) Dead Reckoning: This method uses estimates the current future of the mobile node by using the current position and a speed and direction [92]. For doing this,inertial navigation technology is used. The accelerometers are used to determine the modification of the user position when there exist a detection of acceleration on a certain direction. This measurement is enhanced by combining this information with gyroscopes that can deliver a more accurate estimation on direction changes. When initial position and orientation of the mobile device are known, subsequent positions, orientations and velocities can be updated continuously via Dead Reckoning (DR) without the need for external reference positions. The accuracy of the propagated position depends heavily on the quality of the initially provided position and continuous orientation estimation.

V. Underground Mine Positioning Systems: Current Solutions and Research Directions

In current literature, different systems have been proposed to deal with the problem of underground mine positioning. In this section each proposal is presented and analysed.

In Table II the main contributions existing in literature are presented. These underground mine positioning systems have been evaluated based on their technological principle, topology, infrastructure requirements, accuracy and implementation.

In previous section we have introduced different technologies that could be used for underground environments. In this Section we will discuss their main advantages and

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TABLE II
PROPOSED POSITIONING SYSTEMS FOR UNDERGROUND MINES

Article	Technological principle	System topology	Infrastructure deployment	Estimation method	Reported accuracy	Implementation
[36]	RF Zig-bee	Indirect remote positioning	Strategic deployment	Range based RSSI	2.6m static 4.5m dynamic (average error)	Real scenario
[37]	RF Zig-bee	Self positioning	Strategic deployment	Not specified	Not specified	Computational Simulation
[38]	RF Zig-bee	Remote positioning	Strategic deployment	Range based RSS	Not specified	Real scenario
[39]	RF wireless sensor network	Self positioning	Strategic deployment	Range based RSSI	3.67 m to 6.972 m (average error)	Computational Simulation
[40]	RF Zig-bee	Indirect remote positioning	Strategic deployment	Range based RF-TOF	3 m CDF 86.4%	Real scenario
[41]	RF Zig-bee	Self-positioning	Strategic deployment	Range based RSSI	2 m 100% CDF	Computational Simulation
[42]	RF Zig-bee	Remote positioning	Strategic deployment	Range free Cell ID	Room level	Real scenario
[43], [44]	RF Zig-bee	Remote positioning	Strategic deployment	Not specified	Not specified	Computational simulation
[45]	RF Zig-bee	Remote positioning	Strategic deployment	Range based not specified	Not specified	Not specified
[46]	RF Zig-bee	Self positioning	Strategic deployment	Range based TOF	5 m	Real scenario
[47]	RF not specified	Remote positioning	Strategic deployment	Range free hop-count method	3m normalized root mean square error	Computational Simulation
[48], [49]	RF not specified	Not specified	Strategic deployment	Fingerprint	50 cm controlled environment 1.28 m non-controlled	Real scenario
[50]	RF nanoPAN 5375	Self positioning	Strategic deployment	Range based TOA	$\leq$ 2 m CDF	Real scenario
[51], [52]	RF WLAN	Remote positioning	Pre deployed	Fingerprinting	2 m average error	Real scenario
[53]	RF WLAN	Indirect remote positioning	Pre deployed	Fingerprinting	2-10 m error	Real scenario
[54]	RF WLAN	Indirect remote positioning	Pre deployed	Fingerprinting	3-5 m	Real scenario
[55]	RF WLAN	Indirect remote positioning	Pre deployed	Fingerprinting	$\leq$ 20 m mean error	Real scenario
[56]	RF WLAN	Self positioning	Pre deployed	Fingerprinting	1.6 m	Computational simulation
[57]	RF low power WPAN	Remote positioning	Strategic deployment	Range free hop-count method	Not specified	Computational Simulation
[58]	RF low power WPAN DASH-7	Indirect remote positioning	Strategic deployment	Range Based RSS	2-4 meters	Real scenario
[59]	RF (leaky feeder)	self positioning	Strategic deployment	Range based (RSS or TOA)	Depends on number of connected nodes and user's speed	Computational Simulation
[60]–[62]	RF GSM leaky feeder	Indirect self positioning	Pre deployed	Fingerprint	160 m 90% CDF	Real scenario
[63]	RF UWB	Indirect remote positioning	Strategic deployment	Fingerprint	Not specified-feasibility study	Not specified
[64]	RF UWB	Self positioning	Strategic deployment	Range based TOA	not measured	Computational simulation
[65]	RF UWB	Self positioning	Strategic deployment	Range based TOF	0.3 m CDF	Real scenario
[66]	RF Bluetooth	Indirect self positioning	Strategic deployment	Range free Cell ID	Depends on network deployment	Real scenario
[11]	RFID	Self-positioning	Strategic deployment	Range free	From centimetres to few meters	Real scenario
[67]	Acoustic	Self positioning	Strategic deployment	Range based bilateation AOA and range free proximity supported	25 cm mean absolute error	Real scenario
[68]	Magnetic field	Indirect remote positioning	Strategic deployment	Range based received magnetic field strength	0.767 RMSD	Computational Simulation
[69]	Magnetic field	Self-positioning	Strategic deployment	Range based RSSI	$<$1.2 m several tries	Real scenario
[70]	Hybrid Acoustic/UWB	Indirect self positioning	Strategic deployment	Range based TOF	3 cm CDF	Real scenario

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TABLE II
(Continued.) PROPOSED POSITIONING SYSTEMS FOR UNDERGROUND MINES

Article	Technological principle	System topology	Infrastructure deployment	Estimation method	Reported accuracy	Implementation
[71]	Hybrid UWB/INS	Indirect self positioning	Strategic deployment	Range based TOA	15 cm CDF	Real scenario
[72], [73]	Hybrid RF/INS	Self positioning	Strategic deployment	Range based RF technique/INS particle filter	Not delivered	Computational simulation
[74]	Hybrid INS/image	Indirect self positioning	Strategic deployment	Image positioning and dead reckon	5 m CDF	Real scenario
[75]	Hybrid RFID/INS	Indirect self positioning	Strategic deployment	Fingerprint and INS based method	1.7 m RMS error	Real scenario
[76]	Hybrid INS/scanning laser/RFID	Self positioning	Strategic deployment	INS based method supported by range based RFID	Not specified	Real scenario
[77]	Hybrid RFID/WLAN	Remote positioning	Pre deployed	Cell ID like method	Not specified	Real scenario
[34]	Camera	Self positioning	Strategic deployment	Range based image processing based method	0.381 m curved tunnel and 0.1242 straight tunnel	Real scenario
[78]	Laser	Self positioning	Deployment free	SLAM	Not specified	Real scenario (public dataset)
[79]	Laser	Self positioning	Deployment free	SLAM	Not specified	Real scenario
[80]	VLC	Indirect remote positioning	Pre deployed	Range free Cell ID	Not specified	Real scenario
[81]	VLC	Self positioning	Pre deployed	Range free Cell ID based method	From meter-level to centimetre-level	Computational simulation
[82], [83]	VLC	Self positioning	Pre deployed	Range based RSS trilateration	$\leq$1 m	Computational simulation
[84]	VLC	Self positioning	Pre deployed	Range based RSS trilateration	$\leq$1 m CDF	Computational simulation

disadvantages of this technologies and discuss on their potential usage to fulfill MINERS act. requirements.

When RF technology is used for underground positioning Zig-bee technology have been preferred among the others. Different range based and range free methods can be implemented by using this technology. In particular TOF and RSS based range free methods are preferred since this type of networks are capable of providing signal strength measurements and propagation time measurements without requiring additional hardware. The main problem of using Zig-bee technology to provide localization services is that this type of networks have a low data rate. Due to this, communication applications are constrained to low data-rate demanding services. Some proposals have been evaluated in real scenario reaching level meter accuracy.

On the other hand, WLAN are capable of providing higher data rates. Due to this WLAN architecture can be used as a pre-deployed networks that provides wireless communications with localization capabilities. RSSI based range based and fingerprinting methods are preferred since WLAN devices do not require extra hardware to measure the signal strength. Since WLAN technology are capable to receive and transmit information simultaneoully, they can provide self and remote positioning. The main issue with this type of solution is its limited coverage in underground mine scenario due to propagation characteristics of 2.4 and 5 GHz bands [93]. This constraints will lead to a necessary deployment of a more dense network increasing the required infrastructure.

UWB and Bluetooth technology have been also proposed to face the positioning problem in underground mines. Despite achieving a high accuracy in real scenarios, these techniques are not cost effective. In general UWB devices are expensive and Bluetooth solution requires a high number of devices deployed inside the tunnel to reach acceptable coverage and accuracy.

Finally, an emerging RF-based technology, such as, LPWAN has not been yet tested for positioning in underground mines. Currently, LPWAN has been used mostly in underground mines for environmental monitoring and, positioning capabilities of this technology remain unexamined and the achievable accuracy is not yet defined (nd.).

Acoustic based method proposed by Hammer *et al.* [67] uses an AOA based algorithm for high positioning accuracy. This proposal also support range free localization. The main problem of this proposal is that strategic deployment of infrastructure is required and only self positioning topology is supported. In addition to this, due to the low range coverage of mechanical signals in free space a very dense network must be deployed if the systems intends to cover the entire mining surface. The proposed methods are designed to be

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TABLE III
EVALUATION OF REQUIREMENTS FULFILMENT OF CURRENT POSITIONING TECHNOLOGIES

	Requirements													
Technology	III.A	III.B	III.C	III.D	III.E	III.F	III.G	III.H	III.I	III.J	III.K	III.L	III.M	III.N
RF														
WLAN	✓	✓	✓	✓✗	✓✗	✓	✓	✓	✓	-	✓	✓	✗	✗
RFID	✓	✓	✓	✓	✓	✓	✓	✓	✓	-	✓	✓	✗	✗
WPAN	✓	✓	✓	✗	✓	✓	✓	✓	✓	-	✓	✓	✗	✗
UWB	✓	✓	✗	✗	✓✗	✓	✓	✓	✓	-	✓	✓	✗	✗
LPWAN	✓	✗	✓	✓	✓	✓	✓	✓	nd.	-	✓	✓	✗	✗
Non-RF														
Acoustic	✓	✓	✓	✗	✗	✓	✓	✓	✓	-	✓	✓	✗	✗
Visual	✓	✓	✗	✓	✓	✓	✓	✓	✓	-	✓	✓	✗	✗
Magnetic/ Electrical field	✓	✓	✓	✓	✓	✗	✓✗	✓	✓	-	✓	✓	✗	✗
Visible light	✗	✗	✓	✓✗	✓	✓	✓	✓	✓	-	✗	✓	✗	✗
Hybrid														
Various	✓	✓	✓	✓✗	✓✗	✓	✓	✓	✓	-	✓	✓	✓	✗

robust to typical interference from other sound sources, such as, machinery and conveyor belts among others.

Another technology that has been used for underground mine positioning is magnetic field based localization. Lin *et al.* [68] proposed an indirect remote positioning topology. This proposal is uses a magnetic field RSS based method to estimate the position of mobile node inside the tunnel. This methods shows promising accuracy when the system is evaluated through computational simulation. On the other hand Abrudan *et al.* [69] proposed a magnetic field based positioning which has been implemented in real scenario. The main drawbacks of this proposal is that only self positioning topology is supported and strategic deployment of magnetic beacons is required. Due to this, the coverage of this solution is limited and its scalability is compromised by a higher installation cost. Finally, the method is not robust to magnetic fields fluctuations that could occur in the working face due to regular operation.

Different hybrid positioning methods can be found in literature. Low cost high precision positioning technologies are mainly hybridized with RF technology to support remote positioning or indirect self positioning. As it can be seen INS are used frequently in hybrid approach due to their capability to provide position information with deployment-free infrastructure. Depending on the technology used for position estimation, hybrid technologies may fulfill all the requirements present in underground mines. Particularly, by combining different approaches, the robustness of the system could improve by using the best characteristics of each method and complement their weaknesses with another.

SLAM based approaches are mostly used for inspection of abandoned or unsafe tunnels. These solutions are deployment free in terms of infrastructure. Nevertheless, they require a high number of sensors and computational capacity to perform localization making these approaches more expensive and difficult to implement for day to day operation. For this reason, SLAM technology is not considered as a feasible technology that could provide worker's localization in a regular basis.

Visible light communications based positioning has also being proposed for underground mine environment. This approach uses the already pre-deployed lightning infrastructure and achieves a high accuracy. The main problem of this approach is that they only provide self positioning topology. Furthermore, the overground station is not aware of the position of underground personnel. Range based and range free methods are supported when using VLC for localization purpose. Similarly to the problems encountered in WLAN approach, the coverage of a VLC network cannot always be ensured. There exist some zones inside the tunnel with poor illuminations, thus, making impossible VLC approach to provide positioning. Furthermore, VLC is not robust to many hazards that can be encountered inside the mine. Light signal can be scattered or blocked by dust in suspension and large machinery respectively. Finally, this technology is still in a prototype market maturity stage. Due to this, its current implementation in underground mines is limited.

In Table III we present the main conclusions of this study. Here, each technology has been evaluated based on their capabilities to fulfill the requirements of MINERS act. As it can be seen, LPWAN and hybrid methods show great potential to continue to be evaluated in underground mines. Moreover, there are still many challenges that underground mine positioning technologies should overcome when facing harsh conditions inside the tunnels like the ones encountered in the working face.

VI. CONCLUSION

In this article an in depth analysis of different current localization technologies for underground mine positioning was delivered. An important number of solutions have been proposed after the entry into force of the new MINER act in 2006. Due to the special characteristics of underground tunnels only few approaches can be highlighted as potential solutions capable of meeting the requirements established by this new safety standard. Most of the proposed localization systems achieves the minimum-accuracy requirement ($\leq 60\ m$) shown in Table I. Among all, WLAN, VLC and INS based systems have the advantage of using low or non-required infrastructure to provide position information of personnel inside the tunnel. In particular, WLAN approach has the capability to provide indirect and remote positioning topologies to comply with the requirement of overground monitoring

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along with high speed wireless communications. On the other hand, INS and VLC solutions can only provide self positioning. Due to this, they require to be hybridized with some other wireless communication technology in order to report the position estimation to the overground central monitoring station for emergency response. On the other hand, hybrid systems provides a robust solution to the problems that can be encountered when using a single technology to solve the underground mine positioning problem. As most of the indoor environments, a global positioning solution for underground mine environments has not been yet encountered.

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